

Claude Conversations Export

March – June 2026

Research: SuperPoint · 3D Gaussian Splatting · Temporal Consistency · VIO ·
Thermal Imaging

20 conversations

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1. SuperPoint with 3D Gaussian Splatting for Drone VIO

Date: 2 June 2026 | <https://claude.ai/chat/0be9733d-8621-4ad8-a43d-f020a399bdba>

■ USER

I need to write a state of the art / literature review on my project. I am working on a modification of SuperPoint to use 3D Gaussian Splatting, and to enforce temporal consistency (keep detecting the same keypoints). The goal is to use it on a drone for VIO. Can you find me literature review / state of the art / related works on all the related topics, and help me with an idea of how to structure it?

■ CLAUDE

Researched key areas across SuperPoint, 3DGS, temporal/geometric consistency, and VIO for drones. Proposed a structured literature review covering: (1) Introduction & Motivation, (2) Local Feature Detection (SIFT, ORB, SuperPoint, D2-Net, R2D2, DISK, LightGlue), (3) 3D Scene Representations (NeRF → 3DGS), (4) Temporal Consistency (TILDE, FE-DeTr), (5) 3DGS for Localization (GSplatLoc, GS-SLAM), (6) VIO for Drones (VINS-Mono, MSCKF, OKVIS, EuRoC MAV dataset), and (7) Gap analysis mapping to your contributions. Key papers to cite: DeTone 2018 (SuperPoint), Kerbl 2023 (3DGS), Rakhimov 2024 (GSplatLoc), Yan 2024 (GS-SLAM), Qin 2018 (VINS-Mono), Lindenberger 2023 (LightGlue).

2. InfoNCE vs Hinge Loss for SuperPoint

Date: 2 June 2026 | <https://claude.ai/chat/34f1b4ba-ebb3-4849-83fa-6fea4441f2a7>

■ USER

Explain to me, simply, the pros and cons of using InfoNCE over hinge loss for SuperPoint.

■ CLAUDE

InfoNCE considers all negatives simultaneously (richer gradient, probabilistic interpretation, stable training) but needs large batches. Hinge loss is intuitive with a margin m , works with small batches and hard-negative mining, but saturates and is sensitive to the margin hyperparameter. Use InfoNCE for large batches and stable training; use hinge when batches are small or compute is tight.

■ USER

Here is what I said — is it correct? [submitted LaTeX explanation]

■ CLAUDE

Reviewed the explanation and corrected the LaTeX table presenting temporal consistency evaluation results across SuperPoint, Ours (PrP), and Ours (Temporal). Fixed underlining for best results across tracking metrics (detector/descriptor track length, spatial coverage, parallax), descriptor matching metrics (precision, matching score, match error), and track survival at $N=5/10/25/50$ frames.

3. LaTeX Table for Process Comparison

Date: 2 June 2026 | <https://claude.ai/chat/ccfff4dd-151b-489f-96e0-10456e80abfe>

■ **USER**

Make me a LaTeX table with the main results comparing these 3 processes.

■ **CLAUDE**

Created a LaTeX table comparing Ours (Homography), Ours (PrP), and Ours (Temporal) across tracking metrics, descriptor matching vs GT correspondences, and track survival (N=5/10/25/50 frames). Later also created a second table for MagicLeap dataset results with SuperPoint as baseline. Used booktabs, resizebox, and multirow formatting.

■ **USER**

How can I make these two images go a bit beyond the margins because they look small right now?
[LaTeX figure code]

■ **CLAUDE**

Use `\makebox[\textwidth][c]{}` and increase individual image widths beyond $0.49\text{\textwidth}$. Set both to $\sim 0.55\text{--}0.65\text{\textwidth}$. The `makebox` centers content within `\textwidth` but allows contents wider than the box without layout errors, creating a symmetric bleed into both margins.

4. Paper Explanation: SP-Flow

Date: 31 May 2026 | <https://claude.ai/chat/d14cd22e-0c6a-401a-b909-27b252cc0bdc>

■ USER

Explain this paper. [SP-Flow PDF uploaded]

■ CLAUDE

SP-Flow merges SuperPoint's self-supervised framework with optical flow to enforce temporal consistency across consecutive frames. Training has 3 stages: pre-training on synthetic shapes, self-labeling via homographies on real images, then joint training using Lucas-Kanade optical flow to classify well-tracked vs poorly-tracked keypoints. Plugged into ORB-SLAM2, it outperforms ORB-SLAM2 on TUM RGB-D and KITTI, producing 301 correct matches vs 214 (ORB) and 126 (GCNv2) from the same 1000 keypoints.

■ USER

Does this paper do anything to improve the descriptors too, or is it mainly to improve the detector?

■ CLAUDE

Primarily improves the detector. SP-Flow drops SuperPoint's separate descriptor decoder and uses ORB-compatible descriptors computed in a single forward pass. The matching gains (301 vs 214 vs 126) come from better keypoint placement and temporal consistency — not richer descriptors. The insight is: stable, trackable keypoints improve matching quality even with a simple descriptor.

5. Correction d'accents et de fautes (PDF)

Date: 18 May 2026 | <https://claude.ai/chat/12ffa716-150a-47ad-bd34-e4dd0c69ff8e>

■ USER

Remets les accents et corrige les fautes. [PDF uploaded — synthèse chapitre 5, géographie économique belge]

■ CLAUDE

Corrected all missing accents (é, è, ê, à, â, î, ô, û, ù) and spelling errors including: nécessité, accède, développe la, hyperspécialisées, multiséculaire, peuplée, Woluwe. Also fixed typography: French guillemets (« »), en-dashes for date ranges (1830–1873), curly apostrophes. Regenerated the full document as a corrected PDF using reportlab.

6. Temporal Consistency in Keypoint Detection: TILDE Paper

Date: 16 May 2026 | <https://claude.ai/chat/b8618e1f-3296-4cc3-83c1-25090a1cd84b>

■ USER

Explain this paper: <https://arxiv.org/pdf/1411.4568> — Is it about temporal consistency (i.e. keep detecting the same keypoints in a sequence of images)?

■ CLAUDE

TILDE (Temporally Invariant Learned DEtector) is about repeatability across drastically different imaging conditions — different weather, seasons, time of day — at the same static viewpoint. Not video tracking. Training uses webcam image stacks from AMOS dataset; stable keypoints across all conditions become pseudo-labels. The learned regressor includes a temporal regularizer forcing similar scores at the same locations regardless of illumination. Achieves ~48% repeatability vs ~30% for the next best method on the Webcam dataset.

7. Adapting SuperPoint-PRP Warping for 3DGS

Date: 5 May 2026 | <https://claude.ai/chat/40c102fb-ac01-4104-b30d-8bfe4c782f2d>

■ USER

I am trying to modify SuperPoint-PRP to work with 3DGS instead of NeRF. The warping doesn't work right now. [Code for NeRF warping + 3DGS dataset creation shared]

■ CLAUDE

Identified key convention differences: your 3DGS camera_transforms are 4x4 C2W matrices (not NeRF's OpenGL convention), depth is metric camera-space Z, and 3DGS uses a sign flip after unprojection. Rewrote warp_points_3DGS() with correct pipeline: sample depth at keypoints → unproject via K_{inv} → transform to world via input_c2w → transform to warp camera via $inv(warp_c2w)$ → project via K. All operations batched. Also flagged performance tips: precompute w2c_warp outside the function if calling repeatedly.

8. Detecting Occluded Regions Between 3DGS Scene Images

Date: 5 May 2026 | <https://claude.ai/chat/818124f8-c5f5-4963-8947-5fb7bea56d68>

■ USER

I have images taken from 3DGS scenes with camera_transforms, intrinsics, depth, and labels. I want to compare 2 images from the same scene and highlight parts from image 2 that are occluded in image 1. My code isn't perfect — enhance it please.

■ CLAUDE

Identified issues: occlusion logic was backwards, no bidirectional check, no out-of-bounds handling, no confidence output, fixed depth tolerance, weak visualization. Rewrote with a clean bidirectional pipeline: `classify_frame2_keypoints()` runs backward pass (frame2→frame1) and forward pass (frame1→frame2). Out-of-bounds is its own category (orange). Produced 6 visualization outputs: per-frame status, side-by-side with correspondence lines, depth comparison, occlusion heatmap, and full composed grid. Added CLI args for `rel_tol`, `abs_tol`, and `split`.

9. Différences de Loss entre SuperPoint et SuperPoint-PRP

Date: 4 May 2026 | <https://claude.ai/chat/0ee0445f-9134-4968-a164-8cab0adb3eb>

■ USER

Quelle est la différence dans les loss entre SuperPoint et SuperPoint-PRP?

■ CLAUDE

La loss de détection (L_p) est identique dans les deux — cross-entropy sur pseudo-labels. La différence est dans la génération des pseudo-labels: SuperPoint utilise l'Homographic Adaptation (homographies aléatoires), PrP utilise la Projective Adaptation (20 images NeRF consécutives reprojetées via paramètres réels). Pour la loss de description (L_d), les deux utilisent la hinge loss, mais la correspondance $s_{hwh'w'}$ est définie via homographie H (SuperPoint) vs reprojection 3D $C(M;p)$ (PrP). Le seuil passe de $\epsilon_s=8$ à $\epsilon_s=4$ car la reprojection projective est plus précise que l'approximation homographique.

10. Adding Term Definitions to LaTeX Equations

Date: 4 May 2026 | <https://claude.ai/chat/9f79d2a7-db01-462f-afd5-0b657f400818>

■ USER

I have a LaTeX file explaining documents. I want you to add the definition of every term of each equation used + add the definition of every important term when they are used (Hinge loss, VGG-style, ...).

■ CLAUDE

Enhanced the LaTeX file to 1538 lines with 103 annotation boxes using a new termbox environment. Added definitions for all equation variables (L_p , L_d , total loss, hinge loss, correspondence indicator, Homographic Adaptation, PrP Steps 1 & 2, 3DGS Gaussian, depth rendering). Also added inline definitions for: SIFT, ORB, FAST, VGG-style, ReLU, BatchNorm, cross-entropy, hinge loss, softmax, self-supervised training, homography, RANSAC, NMS, COLMAP, NeRF, 3DGS, spherical harmonics, alpha compositing, and 15+ more technical terms.

11. Adapting SuperPoint for Temporal Consistency

Date: 27 April 2026 | <https://claude.ai/chat/e3e334e3-30c1-47eb-955e-43252c3b291d>

■ USER

I have the code for SuperPoint. I would like to modify it to train temporally consistent point matching. If I already have a dataset of temporally consistent images, what would be the best way to modify SuperPoint? What are the pros and cons of the different architectures?

■ CLAUDE

Presented 6 architectural approaches: (1) Siamese/Contrastive — add temporal loss on top, least invasive; (2) ConvLSTM/GRU bottleneck — true temporal memory, slower training; (3) Cross-Attention/Transformer — explicit inter-frame correspondence, state-of-the-art; (4) Optical Flow Supervision — use LK flow as pseudo-labels, cheap; (5) LightGlue/SuperGlue matcher fine-tuning — drop-in improvement; (6) Hybrid ConvGRU+Attention — best of both. Recommended: Phase 1 Siamese+Warp, Phase 2 add Cross-Attention, Phase 3 distill into ConvGRU for real-time deployment.

12. SuperPoint-PRP Architecture Differences and Weight Compatibility

Date: 22 April 2026 | <https://claude.ai/chat/8843fd74-c157-4643-8312-eb2d57799a12>

■ USER

How are SuperPoint-PRP and SuperPoint different in their architecture? I am trying to use my weights from SuperPoint-PRP with GlueFactory but it is not compatible.

■ CLAUDE

The mathematical architecture is identical (VGG-style encoder → detector + descriptor heads), but module structure and weight key naming differ completely. `rpautrat/SuperPoint` uses flat key names (`conv1a`, `conv1b`, `convPa...`) while SuperPoint-PRP uses nested module names (`encoder.block_1.conv...`). Provided a complete weight remapping script. Also gave a staged fine-tuning strategy: freeze backbone during GRU warmup (Stage 4), unfreeze top blocks at $lr=1e-5$ (Stage 5), full backbone with layer-wise LR decay (Stage 6). Critical detail: set BatchNorm to `eval()` during Stage 4 or gradients will silently degrade the model.

13. Datasets for Feature Trackers and Keypoint Detectors

Date: 24 April 2026 | <https://claude.ai/chat/8056147b-664b-4827-ada5-2448c55e0cba>

■ USER

What are the different kinds of datasets feature trackers and keypoint detectors such as SuperPoint are trained on?

■ CLAUDE

Covered 7 main dataset categories: (1) Synthetic geometry (shapes with exact keypoint labels — SuperPoint's MagicPoint stage); (2) Large-scale natural images with Homographic Adaptation (MS-COCO 2014 for SuperPoint); (3) Benchmark evaluation datasets (HPatches, Aachen Day-Night, InLoc); (4) Outdoor/street datasets (KITTI, Oxford RobotCar); (5) Indoor scene datasets (ScanNet, MegaDepth); (6) Medical/satellite/specialized domains; (7) NeRF/3DGS-rendered datasets (PrP). Also helped refine the project abstract — final version framing the 3DGS fine-tuning approach for VIO without overcommitting to a specific architecture.

14. Training LightGlue on Multi-Scene Image Datasets

Date: 31 March 2026 | <https://claude.ai/chat/473437fb-7843-4885-b5cc-d576d339b5e1>

■ USER

I have a dataset of images from multiple scenes. How to train LightGlue on them? How to structure my data?

■ CLAUDE

Explained that LightGlue is typically fine-tuned, not trained from scratch. Provided two data structure options: MegaDepth-style (COLMAP sparse reconstruction + depth maps) and homography-based (image pairs + H.txt). Full training pipeline: (1) COLMAP reconstruction, (2) pair mining via co-visibility, (3) SuperPoint feature extraction, (4) ground truth generation via epipolar geometry or depth reprojection, (5) YAML config, (6) training with `gluefactory.train`. Recommended curriculum: warmup (easy pairs) → main training → hard negative mining.

15. LightGlue Dataset Structure Requirements

Date: 30 March 2026 | <https://claude.ai/chat/f9ea000a-7015-43d3-b9cd-4d710a0d8a32>

■ USER

How do I need to structure my data when training my own dataset with LightGlue?

<https://github.com/cvg/glue-factory>

■ CLAUDE

Fetches and analyzes the glue-factory codebase. Required data structure: images folder, pairs.txt (one pair per line), cameras.npz (K intrinsics + T_w2cam 4x4 world-to-camera per image). Each `__getitem__` must return view0/view1 dicts with image tensor, K, T_w2cam, and optionally depth. Provided full custom BaseDataset implementation, YAML config template, and training launch commands. Key note: without depth maps, ground truth uses epipolar geometry (slightly less precise than depth-based supervision).

16. 3D Gaussian Splatting Thermal Imaging Datasets

Date: 27 March 2026 | <https://claude.ai/chat/8482dc14-7b08-4d8f-a390-8458fa645fb8>

■ USER

Find me as many 3D Gaussian Splatting datasets using thermal images — even better if they already have a COLMAP/.ply file.

■ CLAUDE

Found ~16 thermal multiview datasets. Best picks with COLMAP/poses included: (1) RGBT-Scenes (ThermalGaussian, ICLR 2025) — 10 scenes, COLMAP included, Google Drive download; (2) TI-NSD (Thermal3D-GS, ECCV 2024) — 20 scenes/6664 frames, COLMAP poses, 6 scenes released; (3) TSDN (ThermalGS, Remote Sensing 2025) — aerial UAV, 5 buildings day/night; (4) FReStereo — 204K stereo pairs, LiDAR SLAM poses + .ply point clouds; (5) MS² — 162K frames RGB/NIR/thermal/LiDAR. Key reference paper: 'Thermal is Always Wild' (arXiv 2603.20448).

17. Thermal Gaussian Splatted Scene Datasets (PLY files)

Date: 25 March 2026 | <https://claude.ai/chat/800c20b7-4134-4d7b-9825-b3ce529a15ec>

■ USER

Can you find me already Gaussian splatted scenes, but done using thermal images? I want to have the PLY of thermal Gaussian splatted scenes.

■ CLAUDE

Surveyed available thermal 3DGS projects. Pre-built PLY files are scarce — most release training framework + raw images. Best options: (1) ThermalGaussian (ICLR 2025) — RGBT-Scenes on Google Drive, run training to get PLY; (2) Thermal3D-GS — pre-trained weights released, exportable to PLY; (3) ThermalGS+TSDN — repo includes points3d.ply in dataset structure. Fastest path to a PLY: use ThermalGaussian repo (most mature, ICLR 2025, standard 3DGS pipeline).

18. Docker Desktop Installation Dependency Error

Date: 24 March 2026 | <https://claude.ai/chat/460d7b8d-ae86-4e19-8f87-d760a4b5689f>

■ USER

Why? [apt install docker-desktop error: Depends: docker-ce-cli but it is not installable]

■ CLAUDE

Docker Desktop requires docker-ce-cli from Docker's official repository, not Ubuntu's default repos. Fix: (1) remove old Docker packages, (2) add Docker's GPG key and apt repository, (3) apt install docker-ce-cli, (4) then install docker-desktop. The issue is that Ubuntu's repos only include the older docker.io package, not docker-ce-cli.

19. Thermal Image Optical Flow with Occlusion Handling

Date: 19 March 2026 | <https://claude.ai/chat/4fe476ad-29d0-4f3f-8f4c-230b0a6a9c17>

■ USER

I have a dataset with thermal images and a Laser Scanner Leica MS60. I want to sample thermal images and use the LiDAR to create optical flow similar to KITTI format, including occlusion. What is the best way to do this?

■ CLAUDE

Designed a full pipeline: (1) sync thermal frames and LiDAR clouds by timestamp, (2) project LiDAR into thermal camera frame via extrinsics calibration, (3) use IP-Basic to densify sparse depth, (4) back-project every pixel to 3D using dense depth + camera intrinsics, (5) apply relative pose T_{cam1_cam0} to warp 3D points, (6) re-project to frame $t+1$ to get dense flow field, (7) occlusion detection via forward-backward depth comparison, (8) write KITTI-format flow PNG ($64 \times uv + 2^{15}$, uint16). Generated complete Python script with argparse CLI. Key fix: back-project full pixel grid (not just LiDAR points) for truly dense flow.

20. Converting Thermal Images to Greyscale

Date: 18 March 2026 | <https://claude.ai/chat/eed48f0b-11f2-4061-abab-fcacdeb5848a>

■ USER

I have a thermal image. How to correctly convert it to greyscale?

■ CLAUDE

Naive RGB-to-greyscale fails because thermal uses false-color palettes where color encodes temperature, not luminosity. Three correct methods: (1) Reverse known colormap via KDTree nearest-neighbor lookup on matplotlib colormap LUT (most accurate); (2) Use Hue channel from HSV (fast approximation — hue correlates with temperature in most palettes); (3) PCA on RGB channels — the first principal component captures the temperature axis. Also enhanced a batch processing script: fixed indentation bug, computed PCA axis once from first image (not per-image), fixed sign convention (sky=cold=dark), switched output to lossless PNG, and tuned CLAHE to clipLimit=3.0/tileGridSize=(16,16) for thermal building images.